SSC Science & Technology I - Preliminary Examination

SET B - ANSWERS

Q1 (A): 1-a, 2-c, 3-b, 4-b, 5-a Q1 (B): 1) Li 2) True 3) G = 6.67 × 10■^{11} N m^2/kg^2 4) Absolute humidity 5) Power = 5 D Q2 (A): 1) Same valence electrons → same valency. 2) Tungsten has high melting point. 3) Pine oil stabilizes froth. Q2 (B): 1) Power ≈ 100 W, Energy = 1 unit 2) Beyond 2F 3) Methanol, Ethanol, Propanol, Butanol 4) Be: 2,2 (Valency 2); O: 2,6 (Valency 2) 5) Telescopes are placed in space to avoid atmospheric disturbance. Q3 (Any Five): 1) $T^2 \propto R^3$ 2) Refraction twice; r = 40°, e = 55° 3) Current divides in parallel 4) Final temperature ≈ 20°C 5) Rainbow: Refraction + Dispersion + TIR 6) Reactive: K, Na, Ca; Moderate: Mg, Al, Zn; Less: Fe, Cu 7) Time ≈ 2 hours 8) Rainbow explanation Q4: Balanced equation: Equal atoms both sides Law: Conservation of mass Unbalanced: Atoms not equal

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SET A - ANSWERS

Q1 (A): 1-a, 2-a, 3-b, 4-a, 5-c Q1 (B): 1) Melting of ice 2) True 3) Group 1: Alkali metals :: Group 17: Halogens 4) Mirage 5) Valency: Combining capacity of atom Q2 (A): 1) Tungsten has high melting point 2) Simple microscope uses convex lens 3) Atomic radius decreases (statement false) Q2 (B): 1) A = Sodium; B = Calcium; More reactive = Sodium 2Na + 2HCl → 2NaCl + H■ 2) Electrical → Mechanical energy 3) Refraction twice; i = e 4) Oxidation; Esterification 5) Planets, satellites, asteroids, comets; Earth has 1 natural satellite Q3: 1) Myopia (concave lens); Hypermetropia (convex lens) 2) Refraction in water (object appears displaced)